

## Research paper

# Studies on the regulatory effect of Peony-Glycyrrhiza Decoction on prolactin hyperactivity and underlying mechanism in hyperprolactinemia rat model



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## HIGHLIGHTS

- The hyperprolactinemia rat model is established successfully.
- The anti-hyperprolactinemia activity of Peony-Glycyrrhiza Decoction is confirmed.
- PGD modulates the expression of DRD<sub>2</sub> and DAT to suppress the prolactin level.
- PGD restores serum progesterone level to suppress the hyper-level of prolactin.
- Peony play a more important role in PGD-mediated anti-hyperprolactinemia effect.

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## ABSTRACT

Clinical trials have demonstrated the beneficial effects of Peony-Glycyrrhiza Decoction (PGD) in alleviating antipsychotic-induced hyperprolactinemia (hyperPRL) in schizophrenic patients. In previous experiment, PGD suppressed prolactin (PRL) level in MMQ cells, involving modulating the expression of D<sub>2</sub> receptor (DRD<sub>2</sub>) and dopamine transporter (DAT). In the present study, hyperPRL female rat model induced by dopamine blocker metoclopramide (MCP) was applied to further confirm the anti-hyperPRL activity of PGD and underlying mechanism. In MCP-induced hyperPRL rats, the elevated serum PRL level was significantly suppressed by either PGD (2.5–10 g/kg) or bromocriptine (BMT) (0.6 mg/kg) administration for 14 days. However, in MCP-induced rats, only PGD restored the under-expressed serum progesterone (P) to control level. Both PGD and BMT administration restore the under-expression of DRD<sub>2</sub>, DAT and TH resulted from MCP in pituitary gland and hypothalamus. Compared to untreated group, hyperPRL animals had a marked reduction on DRD<sub>2</sub> and DAT expression in the arcuate nucleus. PGD (10 g/kg) and BMT (0.6 mg/kg) treatment significant reversed the expression of DRD<sub>2</sub> and DAT. Collectively, the anti-hyperPRL activity of PGD associates with the modulation of dopaminergic neuronal system and the restoration of serum progesterone level. Our finding supports PGD as an effective agent against hyperPRL.

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## 1. Introduction

Hyperprolactinemia (hyperPRL), a clinical hypothalamic-hypophysis disorder, defined as an abnormally high level of prolactin (PRL) in blood ( $\geq 25$  ng/ml for women and  $\geq 20$  ng/ml for men), and is a notable adverse effect of antipsychotic therapy

[10,12]. Antipsychotic-enhanced PRL secretion correlated with the acute and chronic dopamine (DA)-blocking actions [9]. Although DA agonists such as bromocriptine (BMT) are administrated for antipsychotic-induced high PRL [6], the side effects including aggravated psychosis and abnormal involuntary are a greater risk for patients than hyperPRL itself [6]. Identifying alternative therapy for hyperPRL is highly desired.

An herbal medicine formula Paeoniae-Glycyrrhiza Decoction (PGD) made from Paeonia (PR) and Glycyrrhiza radices (GR) has been introduced into the treatment in ovulation disorders,

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menopause and hyperPRL [18–19,23]. In our clinical trial, a great proportion of patients show improvements on risperidone-induced adverse effects associated with hyperPRL after PGD administration [23]. Further *in vitro* and *in vivo* experiments demonstrate that the hyper-level of PRL in MMQ cells and hyperprolactinemia rat model is significantly reduced after PGD administration [22].

These encouraging results led us to further study the mechanisms underlying anti-hyperPRL effects of PGD in hyperprolactinemia rat model. It is considered that pituitary PRL secretion is regulated mainly through the hypothalamic neuro-endocrine dopaminergic system [9]. Tyrosine hydroxylase (TH) is an enzyme that catalyzes the rate-limiting step of tyrosine conversion in the biosynthesis of DA, and is one of the targets of PRL action within the hypothalamus [2,16]. In addition, dopamine transporters (DAT), responsible for re-uptaking of DA within hypothalamus, is also a component in the overall regulation of PRL homeostasis. D<sub>2</sub> receptor (DRD<sub>2</sub>) shows an important effect on short-loop feedback mechanism of PRL regulation, and the robust PRL expression is observed in DRD<sub>2</sub>-null mice [4]. Moreover, progesterone (P) stimulates the production of deciduas PRL, conversely, it appears to be a potent inhibitor of myometrial PRL production [11].

In the present study, we examined the effects of PGD on the concentration of serum PRL and progesterone. The expression of DRD<sub>2</sub>, DAT and TH in pituitary gland and hypothalamus after PGD administration were determined. Our results revealed that PGD displayed anti-hyperPRL effects not only through the regulation of dopamine mediators, but also through the normalization of other sexual hormone dysfunction, including progesterone.

## 2. Materials and methods

### 2.1. Herbal preparation

The raw materials of Paeoniae and Glycyrrhiza radices that constitute the PGD formula were supplied by the Pharmacy of School of Chinese Medicine in the University of Hong Kong (HKU). The extraction was conducted in an automatic boiling machine. Sliced, broiled Paeoniae and Glycyrrhiza radices (50 g each) were immersed and boiled in a 10-fold volume of distilled water for 2 h. This process was repeated twice as previously reported [22]. The extractive solution was pooled, concentrated, and freeze-dried into a ratio of 1:5 in weight of powder to raw materials for the herbal preparation.

### 2.2. HyperPRL rat model establishment and drug administration

Experimental protocol was approved by Committee on the Use of Live Animals in Teaching and Research (CULATR) of Li Kang Shing Faculty of Medicine of the University of Hong Kong. Sprague-Dawley female rats aged 11 weeks old and weighing 280–300 g were used in the experiment. Animals were housed in groups of two in clear plastic cages and maintained on a 12-h light/dark cycle (lights on 07:00–19:00 h) at 23 ± 1 °C with water and food available *ad libitum*. Vaginal smears were examined daily to ensure that collection of blood samples were conducted in animals in the same stage of the oestrous cycle.

Experimental hyperPRL rat model was produced similarly as previous studies [22]. 45 animals were given intraperitoneally (i.p.) metoclopramide (MCP; Sigma, USA) (50 mg/kg, three times per day), a dopamine inhibitor for 10 days. Another 9 rats were treated with 0.9% salt solution to serve as controls. At day 10, blood was drawn from the tail vein for PRL measurement. Compared to non-treated rats, all 30 MCP-treated animals displayed an at least 80% elevation of serum PRL, and all of them were used for experimental treatment and received gastric PGD at a dose of 0 g/kg, 2.5 g/kg, 5 g/kg, or 10 g/kg, or with 0.6 mg/kg BMT for 14 days in a random

manner ( $n=9/\text{group}$ ). During the experiment, the bodyweight of each rat was monitored everyday. At the end of PGD administration, blood samples were collected and used to determine the concentration of PRL and progesterone. Tissues of pituitary gland and hypothalamus in each group were collected and used to detect the alteration of DRD<sub>2</sub>, DAT and TH expression.

### 2.3. Biochemical analyses

#### 2.3.1. Hormone assay

The concentration of progesterone in rat sera was determined via Elecsys 2010 (Roche, Switzerland) directly. Serum PRL concentration was measured using enzyme-linked immunosorbent assay (ELISA) (Calbiotech, USA).

#### 2.3.2. Western blotting

Western blotting was conducted to determine the expression of DRD<sub>2</sub>, DAT and TH in tissues of pituitary gland and hypothalamus. Tissue proteins were extracted by RIPA buffer (Sigma–Aldrich, USA) containing the 1% protease inhibitor cocktail (Sigma–Aldrich, USA) and 2% phenylmethanesulfonyl fluoride (PMSF; Sigma–Aldrich, USA). The concentration was determined via Bradford protein assay using coomassie brilliant blue G-250 (Bio-rad Laboratories Inc., USA). Proteins were separated by a 10% SDS-PAGE gel and transferred electrophoretically onto nitrocellulose membranes (0.45 µm, Bio Basic, Inc.). Immunodetection was performed with the primary antibodies against DRD<sub>2</sub>, DAT, TH and glyceraldehyde 3-phosphate dehydrogenase (GAPDH) (Santa cruz Biotechnology, Inc., USA) at a dilution of 1:1000 at 4 °C overnight, followed by co-incubation with horseradish peroxidase (HRP)-conjugated secondary antibodies for 30 min at room temperature. The chemiluminescence was determined using ECL detection kits (GE Healthcare, UK). The intensity of protein bands was quantified by scanning densitometry with Quantity One 4.5.0 software.

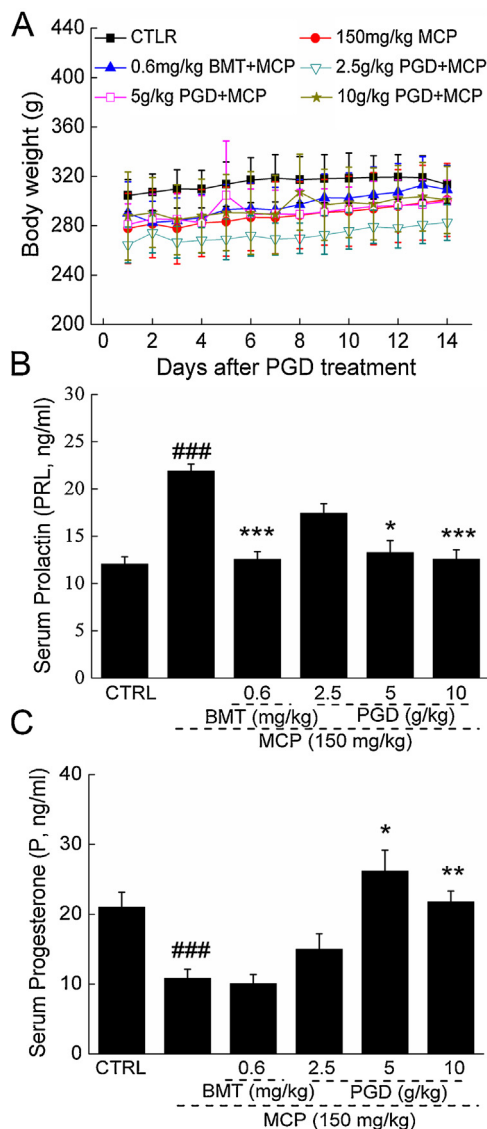
#### 2.3.3. Immunofluorescent examination of brain tissues

After 14-day 10 g/kg PGD and 0.6 mg/kg BMT administration, rats were anesthetized with Ketamine (100 mg/kg) and Xylazine (5 mg/kg), and then transcardially perfused with cold PBS followed by 4% paraformaldehyde for 30 min. Brains were harvested and soaked into 4% paraformaldehyde for 24 h. Further, samples were dehydrated via soaking in 30% sucrose for 48 h. This step was repeated for three times. After dehydration, samples were soaked into TOC in suitable containers and frozen at –80 °C. The slice of brain tissue is cut at 15 µm.

Immunofluorescence was performed to detect the expression changes of DAT and DRD<sub>2</sub> in arcuate nucleus of hypothalamic. The prepared sections of brain were permeabilized with 0.15% Triton X-100 (Sigma–Aldrich, USA) for 10 min, and then incubated with blocking buffer containing 5% goat serum and 0.1% Triton X-100 for 2 h at room temperature. Then the sections were incubated with primary antibodies including anti-DRD<sub>2</sub> and anti-DAT (1:200, Santa Cruz, USA) for 48 h at 4 °C followed with three washes with PBS. Next, sections were incubated with Alexa Fluor 568 (1:500; Invitrogen, USA) and Alexa Fluor 488 (1:1000; Invitrogen, USA) for 2 h at room temperature. Nuclei were counterstained with DAPI (100 ng/ml, Sigma Aldrich, USA) for 15 min at room temperature. Fluorescent images were obtained using a fluorescent microscope (10X; CCD camera, Axio Observer Z1; Carl Zeiss, Germany). Image J software was used to calculate the fluorescence intensity of each picture.

### 2.4. Statistical analysis

A One-Way variance analysis (ANOVA) was used to detect statistical significance, followed by *post-hoc* multiple comparisons



**Fig. 1.** Rat model of hyperPRL was produced by repeated injection with metoclopramide (MCP, 150 mg/kg) for 10 days, followed by treatment with or without bromocriptine (BMT) or PGD at various doses for 14 days. 0.9% salt solution treated rats were served as control (CTRL). Rat bodyweight changes (A), serum PRL (B), and progesterone (C) were measured. Data are expressed as mean  $\pm$  S.E.D. ( $n=9$ ) and analyzed using one-way ANOVA. ### $P<0.001$  vs. controls, \* $P<0.05$ , \*\* $P<0.01$  and \*\*\* $P<0.001$  vs. MCP only.

(Dunn's test). Data are expressed as mean  $\pm$  S.E.M. Statistical significance was defined as  $P<0.05$ .

### 3. Results

#### 3.1. The effects of PGD and BMT on the related hormones in rat model of hyperPRL

PGD with doses of 2.5–10 g/kg or 0.6 mg/kg BMT were given to hyperPRL rats for 14 days. Animal bodyweight was unchanged over a period of treatment with either PGD or BMT (Fig. 1A). The significant differences were observed in response to serum concentration of PRL ( $P=0.011$ ) and progesterone ( $P=0.018$ ) treatment (Fig. 1B and C). For PRL, MCP chronic treatment resulted in an 88.1% elevation of PRL level compared to control group. PGD at doses of 5 g/kg and 10 g/kg and 0.6 mg/kg BMT significantly suppressed MCP-induced PRL over-expression ( $P<0.05$ ), with a suppression

by 44.5% in 5 g/kg PGD, 50.1% in 10 g/kg PGD and 48.5% in BMT compared to MCP alone (Fig. 1B). Serum progesterone levels were significantly lower in MCP alone and 0.6 mg/kg BMT groups compared to non-treated rats ( $P<0.001$ ), but MCP plus 5 g/kg and 10 g/kg PGD groups displayed a significantly elevated progesterone level compared to MCP alone and BMT groups ( $P<0.05$ ) (Fig. 1C). These results suggest that progesterone may be involved in PGD mediated benefits in hyperPRL rat model.

#### 3.2. The effects of PGD and BMT on the expression of dopamine mediators in pituitary gland of hyperPRL rat model

Chronic MCP administration resulted in a 19% reduction of DRD<sub>2</sub> level, a 37% suppression of DAT expression, and a 18% reduction of TH expression in pituitary gland ( $P<0.05$ ) (Fig. 2). After 0.6 mg/kg BMT or 2.5 g/kg, 5 g/kg and 10 g/kg PGD administration for 14 days, DRD<sub>2</sub> levels in pituitary gland were enhanced 13%, 15%, 20% and 19%, respectively ( $P<0.05$ ) (Fig. 2B). For DAT level, 0.6 mg/kg BMT or 2.5–10 g/kg PGD increased the expression of DAT up to 37% or 24–47% respectively ( $P<0.05$ ) (Fig. 2C). In addition, after BMT or PGD treatment, TH levels in pituitary gland were enhanced nearly 34% or 30–36% respectively compared to MCP-treated only rats ( $P<0.05$ ) (Fig. 2D). These data further confirmed that the suppressive effects on hyperactive PRL were related with the modulation of dopamine mediators expression in pituitary gland which were boosted by PGD and BMT administration.

#### 3.3. The effects of PGD and BMT on the expression of dopamine mediators in hypothalamus of hyperPRL rat model

Moreover, the significant alterations among experimental groups in the expression of DRD<sub>2</sub>, DAT and TH were observed ( $P<0.05$ ; Fig. 3). For the expression of DRD<sub>2</sub>, MCP chronic treatment resulted in a 20% reduction of DRD<sub>2</sub> level compared to control rats. PGD and BMT significantly enhanced MCP-suppressed DRD<sub>2</sub> levels nearly 25% and 30%, respectively ( $P<0.05$ ) (Fig. 3B). The expression of DAT and TH was remarkably lower in MCP-treated model rats, which was significantly enhanced after PGD and BMT administration ( $P<0.05$ ) (Fig. 3C and D). These data further confirmed that PGD-mediated suppression on hyperactive PRL was associated with the modulation of dopamine mediators in hypothalamus.

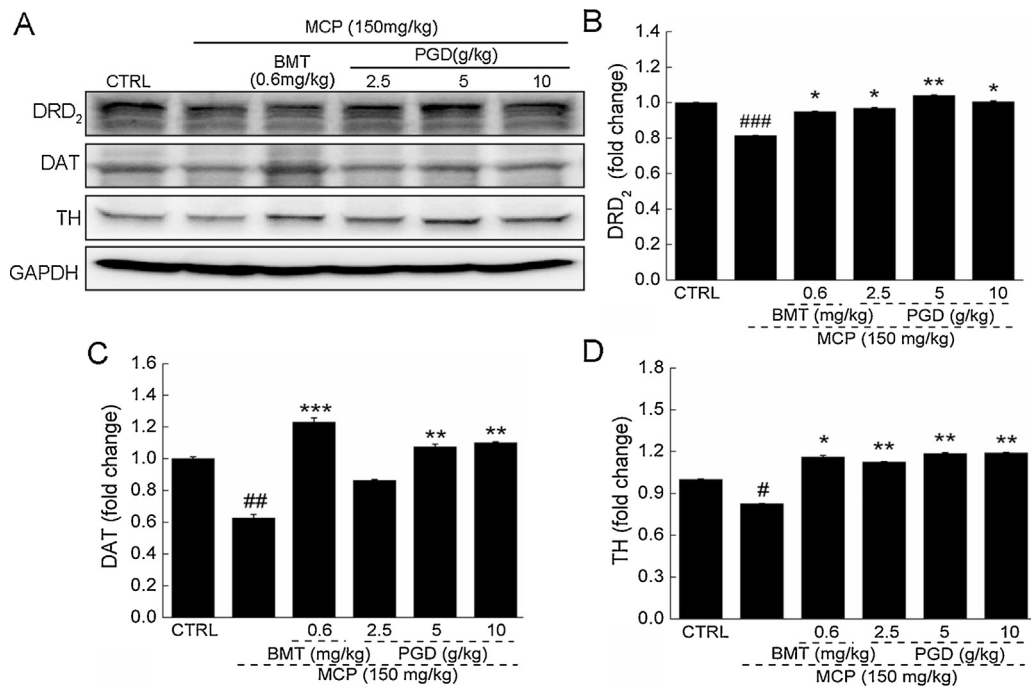
#### 3.4. The effects of PGD and BMT on D<sub>2</sub> receptor and DAT in the arcuate nucleus of the hypothalamus

At the completion of treatment, animal brain were removed and sectioned for immunofluorescent staining to examine the expression of D<sub>2</sub> receptor (green fluorescence) and DAT (red fluorescence) in the arcuate nucleus of the hypothalamus. Compared to non-treated group, hyperPRL animals had a marked reduction of D<sub>2</sub> receptor and DAT expression in the arcuate nucleus; however 10 g/kg PGD and 0.6 mg/kg BMT treatment significantly reversed the level of D<sub>2</sub> receptor (18% and 19% increment vs. MCP only) and DAT (68% and 37% increment vs. MCP only) ( $P<0.05$ ) (Fig. 4).

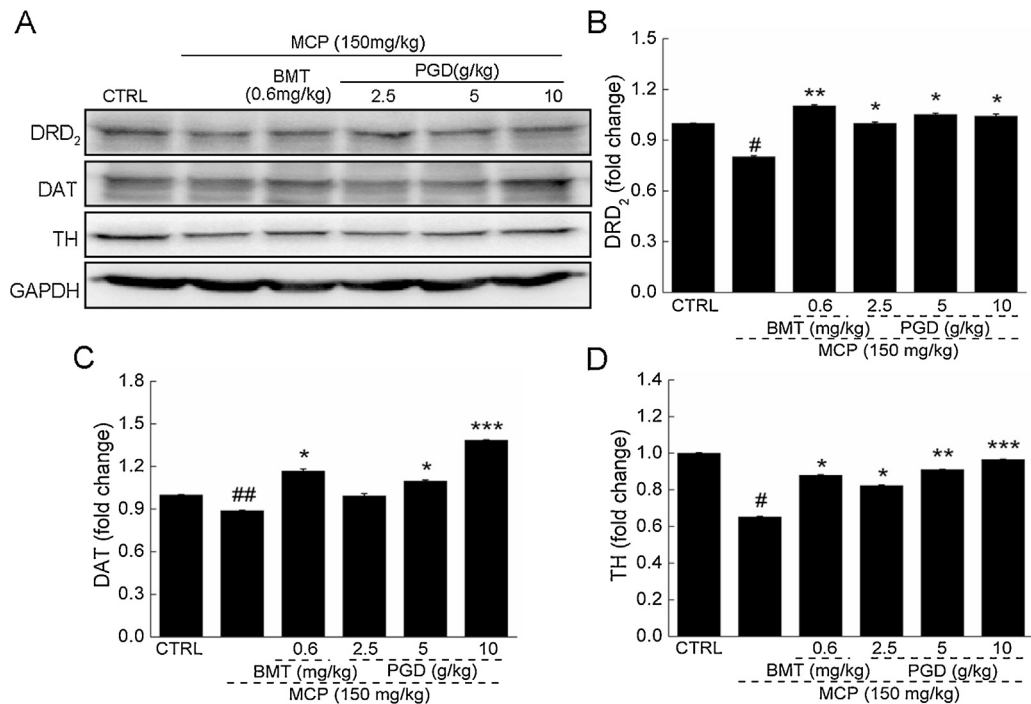
### 4. Discussion

In clinical trial, a great proportion of patients showed improvements on risperidone-induced adverse effects associated with hyperPRL during PGD administration [23]. *In vitro* experiments demonstrated that PGD suppressed PRL release and synthesis in MMQ cells mediated by dopamine system [22]. In the present study, *in vivo* experiment was further performed in hyperPRL female rat model.

As described previously, DRD<sub>2</sub> contributes to PRL regulation in short-loop feed back mechanism [5]. The increment of serum



**Fig. 2.** The effects of PGD and BMT on the expression of dopamine mediators in pituitary gland of hyperPRL rat model. Rat model of hyperPRL was produced by repeated injection with 150 mg/kg MCP, for 10 days, followed by treatment with or without 0.6 mg/kg BMT or 2.5–10 g/kg PGD for 14 days. Non-treated rats were served as control. At the completion of experimental treatment, tissues of pituitary gland of rats in each group were collected respectively. The expression of dopamine mediators including DRD<sub>2</sub>, DAT, TH and PRL was determined by western blot (A). Quantification data of DRD<sub>2</sub> (B), DAT (C) and TH (D) were normalized by GAPDH respectively. Data are expressed as mean  $\pm$  S.E.D. ( $n=6$ ) and analyzed using one-way ANOVA. # $P<0.05$ , ## $P<0.01$  and ### $P<0.001$  vs. control group, \* $P<0.05$ , \*\* $P<0.01$  and \*\*\* $P<0.001$  vs. MCP only.

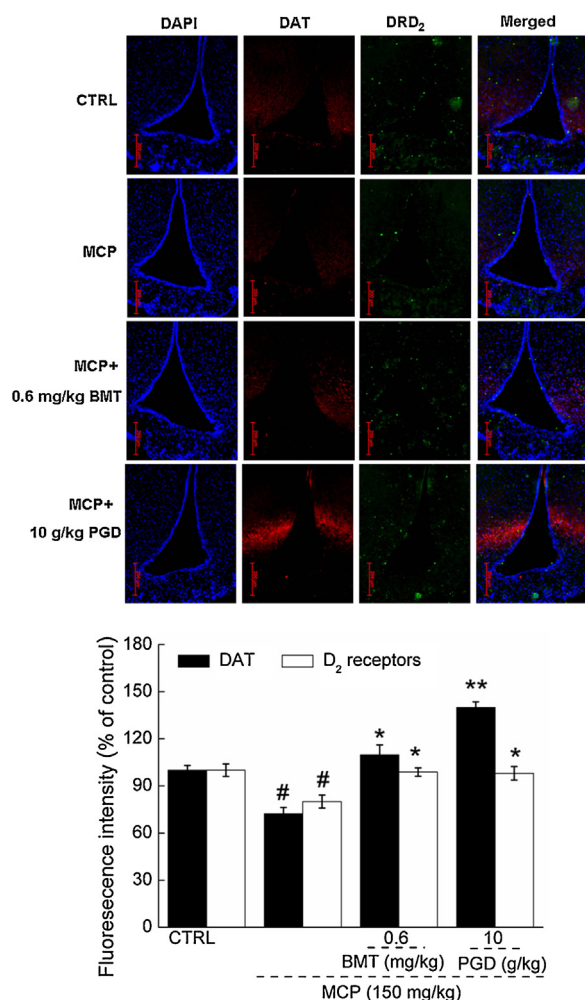


**Fig. 3.** The effects of PGD and BMT on the expression of dopamine mediators in hypothalamus of hyperPRL rat model. After 14 days treatment, tissues of hypothalamus of rats in each group were collected respectively. The alteration of DRD<sub>2</sub>, DAT, TH and PRL level in hypothalamus was determined by western blot (A). Quantification data of DRD<sub>2</sub> (B), DAT (C) and TH (D) were normalized by GAPDH respectively. Data are expressed as mean  $\pm$  S.E.D. ( $n=6$ ) and analyzed using one-way ANOVA. # $P<0.05$  and ## $P<0.01$  vs. controls, \* $P<0.05$ , \*\* $P<0.01$  and \*\*\* $P<0.001$  vs. MCP only.

PRL levels were observed in DRD<sub>2</sub>-null mice [4]. Amount of studies revealed that administration of antipsychotic medications are responsible for PRL levels enhancement [13,21]. The abnormally high level of PRL during antipsychotic treatment is related to the

blockade of DRD<sub>2</sub>, and most of antipsychotic medications are considered as potent DRD<sub>2</sub> antagonist [15]. The effect of PGD on DRD<sub>2</sub> expression is studied in PC12 cells which are widely used for determination of dopaminergic system [20]. Our present results showed





**Fig. 4.** Immunofluorescence was used to detect the expression of D<sub>2</sub> receptor (green) and DAT (red) in the arcuate nucleus. The nucleus boundary was marked using diamidino-2-phenylindole (DAPI, blue) (10× magnification; Scale bar: 200 μm). After hyperPRL rat model established successfully, hyperPRL rats were administrated with or without 0.6 mg/kg BMT or 10 g/kg PGD. Non-treated rats were served as control. Image J software was used to calculate the fluorescence intensity of each picture. Data are expressed as mean ± S.E.D. ( $n=3$ ) and analyzed using one-way ANOVA. <sup>#</sup> $P<0.05$  vs. controls, <sup>\*</sup> $P<0.05$  and <sup>\*\*</sup> $P<0.01$  vs. MCP only. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

that administration of BMT and PGD enhanced the expression of DRD<sub>2</sub> in hypothalamus and pituitary gland indicating PGD is a therapeutic agent for antipsychotic medications-induced hyperPRL. In addition, TH, an enzyme catalyzing the conversion of tyrosine, is one of the targets for PRL activation within hypothalamus [2,16]. It is reported that oestrogens, responsible for higher basal serum PRL level, suppresses basal TH activity, and reduces TH mRNA levels [1]. Long-term MCP treatment caused a reduction of TH expression in hypothalamus and pituitary gland; meanwhile, treatment with BMT and PGD restored the expression of TH significantly. Since TH is the rate-limiting enzyme in the biosynthesis of DA, PGD may indirectly regulate DA concentration. Furthermore, BMT and PGD strongly enhanced the expression of DAT in hypothalamus and pituitary gland. The expression and secretion of PRL gene are reduced by the blockade of DAT in female rats further confirming that DAT participates in the physiological regulation of PRL [8]. It can be explained that an increase of DAT expression results in an inhibition of DAT release to the synaptic gap. Collectively, the modulation of dopaminergic neuronal system of PGD contributes to its anti-hyperPRL effect which is similar as BMT.

Furthermore, PRL production and negative feedback regulation are largely controlled by tuberoinfundibular dopaminergic (TIDA) neurons, which terminate around the capillary loops in the arcuate nucleus of the hypothalamus [5]. Dopamine, that is released from TIDA and delivered into anterior lobe, is considered the major physiological regulator of PRL secretion [14]. We showed significant increment of DAT and D<sub>2</sub> receptor expression in the arcuate nucleus of the hypothalamus in hyperPRL rats chronically treated with BMT and PGD, suggesting that, like conventional anti-hyperPRL agent, the anti-hyperPRL effect of PGD is associated with the modulation of DAT and D<sub>2</sub> receptor function in neuroendocrine dopaminergic neuronal pathway by directly enhancing TIDA activity.

After BMT and PGD administration, the reduction of serum PRL level was observed; interestingly, only PGD restored the blood progesterone compared to control level. This effect is very different from BMT which showed no effects on progesterone level in serum. Antipsychotic-induced hyperPRL-like symptoms are related to dysfunction of sexual hormones [12]. Our findings suggest that sexual hormone function contributes to PGD-mediated anti-hyperPRL effects. Previous studies provide sufficient evidence to support this assumption, and it is revealed that PGD possesses regulation effects on sexual hormones in women suffered with abnormal menstruation [19]. Additionally, compounds separated from PGD directly regulates PRL and testosterone production in animal experiments and clinical trials [3,17].

In conclusion, we confirmed the effectively suppression effects of PGD on hyper-level of PRL *in vivo* experiments. Our results further demonstrated that PGD showed anti-hyperPRL effects associated with multiple factors including regulation dopamine mediators and normalization of serum progesterone level. Our finding provides experimental evidence of PGD to be used as an effective agent against hyperPRL.

## 5. Conflict of interest

The authors have declared that there is no conflict of interest.

## Acknowledgements

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## Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.neulet.2015.08.024>.

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